A NEW TECHNIQUE FOR VISUALIZATION OF FOREST FIRE SMOKE PLUMES USING MODIS DATA

Izumi Nagatani[1], *Jun-ichi Kudoh*[1], *Koichi Kawano*[2]

[1]GSIS Tohoku University
[2]Tohoku Institute of Technology

ABSTRACT

Forest fire smoke detection by satellites is important and required for monitoring air pollution and human health. MODIS smoke detection algorithms are under development. The common problem is to separate smoke from clouds. To overcome this issue we propose a new visualization technique of a false-color composite image that composed of Smoke Reflectance Index (SARI), MODIS channel 7 reflectance, and Water Index (WI). The SARI and WI were developed in this study. The false-color composite image shows smoke in reddish and clouds in pink-white. Smoke pixels are easily identified and sampled. Overall smoke pixels are detected by their training dataset.

In this paper, we present a case study of Russia and Mongolian forest fire in 2009. The result of smoke detection was compared to those of existing method. It was confirmed that the proposed method detected smoke pixels more accurately.

***Index Terms*—** forest fire smoke, smoke plume detection, visualization, MODIS, image enhancement

1. INTRODUCTION

Forest fire and smoke plume detection is important application of remote sensing technology [1]. Weather satellites data such as NOAA/AVHRR and DMSP/OLS, which are high frequent observation systems at a same earth location, have been used well for this purpose. After emergence of the Earth Observation System (EOS) mission, Terra and Aqua MODIS measurements are used as a primary data.

The Moderate-resolution Imaging Spectroradiometer (MODIS) aboard the Terra and Aqua satellites observe the earth surface with 36 channels of electromagnetic wave from visible (VIS) to thermal (TIR) bands and is suitable for forest fire and smoke detection. A Forest fire detection algorithm (MOD14) was developed by NASA science team and MODIS fire products have been produced operationally. Their temporal dataset is commonly used as a standard global fire product.

MODIS smoke detection algorithms are under development. Xie et al. was developed a multi-threshold algorithm for detecting smoke pixels from MODIS data [2]. Though it is difficult to separate smoke from clouds precisely, in case of emergency, smoke information of satellites is critical and required for monitoring air pollution and human health.

In this paper, we propose a new visualization technique for forest fire smoke identification especially for precise discrimination between smoke and clouds. The initial concept of this method is to identify smoke pixels on a false-color composite image that consist of three indices of smoke aerosol reflectance, mid-infrared reflectance and water vapor level. In order to prepare those three data, we developed three reflectance indices that are Smoke Aerosol Reflectance Index (SARI), Land Water Reflectance Index (LWRI) and Water Index (WI). The SARI shows visible-band reflection by using Aerosol Enhancement technique (AE), which is a function of MODIS channel 1 (red) and channel 3 (blue), and has a linear correlation with Aerosol Optical Thickness (AOT) at 0.55μm over land. The LWRI is derived from MODIS channel 2 (NIR) and channel 7 (SWIR) and used to create Water Index (WI) together with Aerosol Vapor Index (AVI). Thus, the proposed false-color image is created from the SARI, MOD7 and WI.

The method was applied to four MODIS scenes of large forest fires in North Korea, Russia, Australia and Mongolia for case studies. It was confirmed that the smoke pixels could be detected by the three indices, SARI, MOD7 and WI and their false-color composite image (R:SARI, G:MOD7, B:WI) was useful for discriminating smoke pixels from any clouds. In this paper, we present a case study of Russia and Mongolian forest fire.

2. METHODOLOGY

2.1. Smoke aerosol reflectance index

MODIS reflectance measurements at the top of atmosphere (TOA) basically consist of earth surface reflectance, aerosol reflectance and Rayleigh scattering. The Rayleigh scattering is removed by atmospheric correction procedure. The smoke aerosol reflectance is retrieved from difference between

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MODIS channel 3 and channel 1 with weakening earth surface reflectance. MODIS channel 3 is sensitive to smoke and haze but MODIS channel 1 is less sensitive. Thus, the difference between MODIS channel 3 and channel 1 emphasize aerosol reflectance and Aerosol Enhancement function (AE) is defined as equation (1).

$$AE = 2.0 * MOD3 - (MOD1 - Cm) \quad (1)$$

Where MOD1 and MOD3 is atmosphere corrected reflectance of MODIS channel 1 and channel 3, respectively. Cm is constant value of minimum reflectance of MODIS channel 1 over dark vegetation and 0.025 is set here.

The AE is useful function for showing smoke aerosol existence but it does not represent smoke intensity and cannot discriminate smoke plumes from other haze. Thus, we modified it to Smoke Aerosol Reflectance Index (SARI) as equation (2) which has linearly correlation with Aerosol Optical Thickness (AOT) at 0.55μm.

$$SARI = \alpha * AE$$
$$\alpha = \exp(AE/0.35) \quad \text{(over land)} \quad (2)$$
$$\alpha = 1.0 \quad \text{(over ocean)}$$

2.2. LWRI and WI

Aerosol Vapor Index (AVI), which is brightness temperature difference between MODIS channel 31 and channel 32 (equation 3), is used for atmospheric water vapor detection [3].

$$AVI = BT31 - BT32 \quad (3)$$

Where the BT31 and the BT32 is MODIS channel 31 brightness temperature and those of channel 32, respectively.

The Land Water Reflectance Index (LWRI) was developed for detecting whitish water such as warm clouds, ice and snow. The index is defined as equation (4).

$$LWRI = 2.0 * MOD2 - MOD7 - NDVI \quad (4)$$

Where MOD2 and MOD7 is atmosphere corrected reflectance of MODIS channel 2 and channel 7, respectively. The NDVI, which is Normalized Difference Vegetation Index derived from MODIS channel 1 and channel 2, is used for subtracting effects of vegetation water content.

The total Water Index (WI) is calculated by equation (5).

$$WI = 0.5 * AVI + LWRI \quad (5)$$

2.3. False-color composite image

A false-color composite image, which is R: SARI, G: MOD7, B: WI, is created. The input-range of each component is set as SARI: 0.0-0.3, MOD07:0.0-0.8 and WI:0.0-1.0.

By image interpretation, smoke pixels are identified and their SARI, MOD7 and WI are sampled as training dataset. After that, the total smoke pixels are detected by 80% of probability density of Gaussian distribution function and SARI>0.2

3. CASE STUDY

Large forest fires in Russia and Mongolia border occurred in 2009. Both Terra and Aqua MODIS captured the forest fires. The case study of this paper used Aqua MODIS data (May 21, 2009 05:20 UTC) in which a number of heavy fire smoke was appeared around Lake Baykal. The location of study site is shown in Figure 1.

The Aqua/MODIS Level 1b products were collected from the NASA LAADS website and atmospheric correction and brightness temperature conversion were performed by using CREFL and MOD14 software which were distributed by NASA Direct Readout Laboratory. And then, their SARI, LWRI and WI images were created.

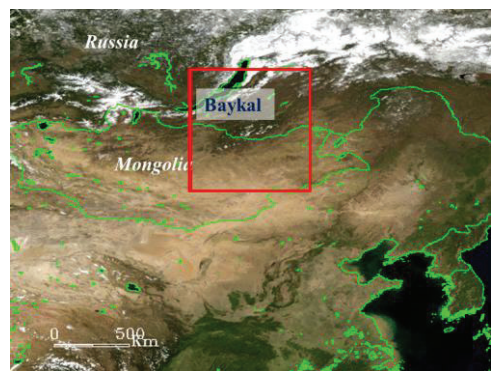

Figure 1. Location of study site (in red square)

Table 1. MODIS data used in the case studies

	Study site	Date of data (UTC)	Satellite
S1	Russia and Mongolia	May 21, 2009 05:20	Aqua
S2	North Korea	Apr. 13, 2011 04:50	Aqua
S3	Australia	Sep. 26, 2011 03:50	Aqua
S4	Russia and China	Oct. 8, 2011 04:35	Aqua

The true color image (Figure 2) shows a number of heavy smoke plumes in gray-white and clouds in white. Soil is dark brown and forest is very dark or black. Large smokes are identifiable but small smokes are obscure.

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Figure 3 shows the SARI image in which smoke plumes are shown in yellow. This pseudo-color image was useful for smoke interpretation. Brown color shows non smoke, gray color shows light smoke or haze, yellow color shows high dense smoke. Clouds were masked with black by using thresholds BT31<273K or LWRI>0.4. Though this image is useful for understanding smoke condition, the cloud mask is not accurate and causes misidentification.

Figure 4 shows the proposed false-color composite image. Smoke, clouds and lands are identifiable in different color. Smoke is shown in reddish, clouds are in pink-white and lands are in greenish. Smoke pixels were identified and their SARI, MOD7 and WI were sampled as training dataset (about 30,000 pixels). Their averages and standard deviations are shown in Table 2. In comparison to other site such as North Korea or Australia, the SARI of this site (S1) is very high and moderately high in WI. This suggests that the smoke had higher amount of vapor.

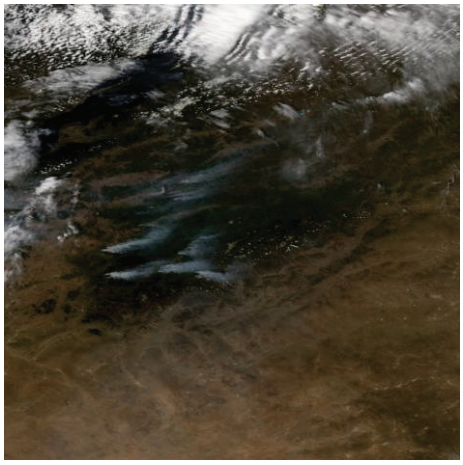

Figure 2. MODIS true color image

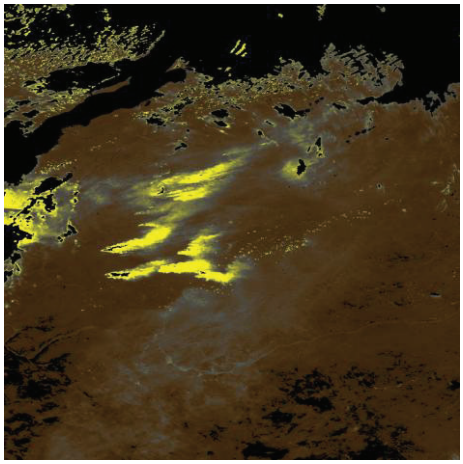

Figure 3. SARI image

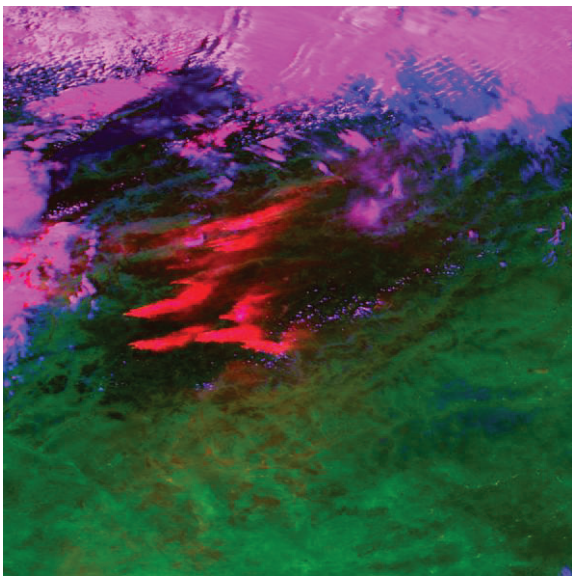

Figure 4. False-color composite image

Table 2. Statistics of smoke pixels

	SARI		MOD7		WI	
Site	Mean	STD	Mean	STD	Mean	STD
S1	0.597	0.206	0.086	0.031	0.252	0.134
S2	0.337	0.081	0.205	0.045	0.029	0.097
S3	0.388	0.081	0.167	0.047	0.262	0.091
S4	0.343	0.059	0.171	0.030	0.171	0.072

4. VALIDATION

Since there is no appropriate validation method for false-color composite image, the smoke detection image was compared to those of existing method that was multi-threshold method developed by Xie et al [2]. Though the multi-threshold method is useful for automatic detection, the accurate thresholds are required and not easy to be defined. In our proposed method, the smoke pixels are identified easily and accurately.

Figure 5(a) and 5(b) shows the result of multi-threshold method and those of proposed method, respectively. In these figures the red dots represent detected smoke pixels. The overall accuracy of these results seems same.

Figure 6 shows differences between the result of multi-threshold method (a) and those of proposed method (b). The white ellipse show whitish clouds and misidentified smoke pixels in red dots. While the yellow circles show undetected smoke pixels in gray and smoke in red. The proposed method results less number of misidentified pixels and

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undetected pixels than the multi-threshold method. Consequently, the proposed method, which was using false-color composite image, detected smoke pixels accurately.

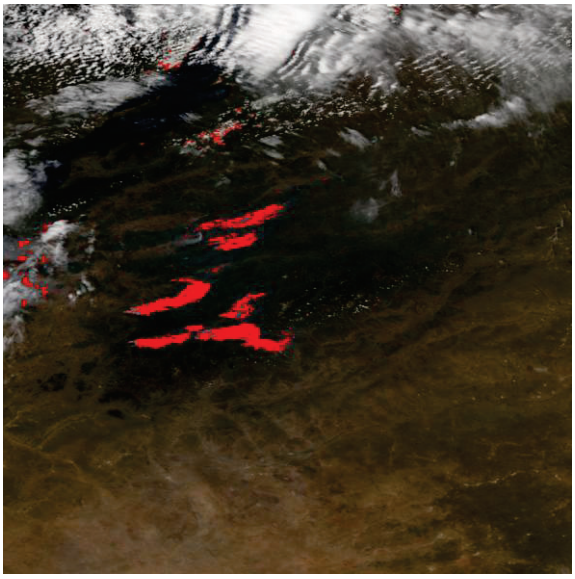


Figure 5(a). Smoke detection by multi-threshold method

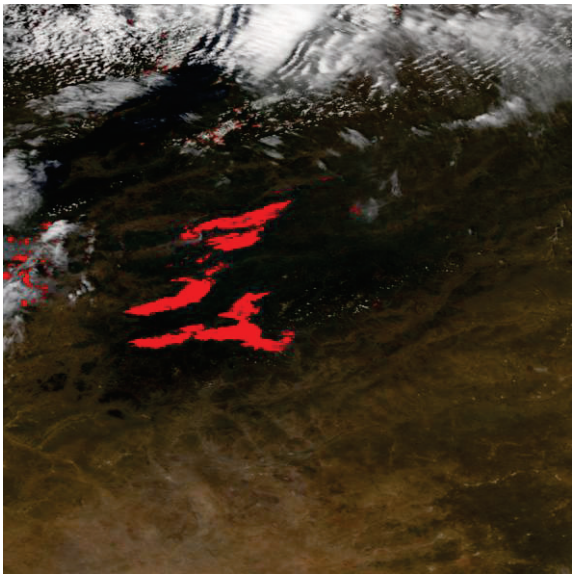
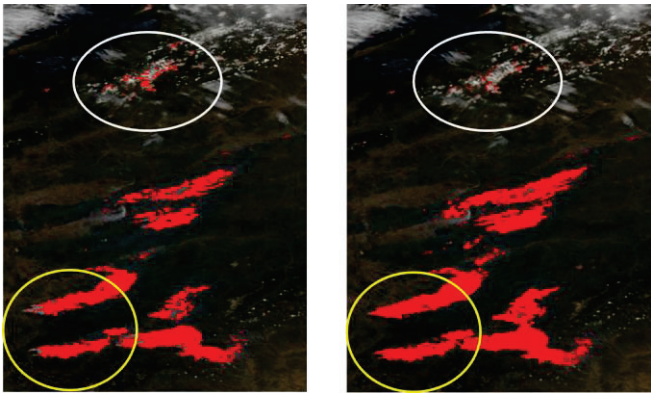


Figure 5(b). Smoke detection by proposed method



(a) Multi-threshold method (b) Proposed method

Figure 6. Comparison of the smoke detection results

5. CONCLUSION

We developed a new technique for visualization of forest fire smoke plume using MODIS data. A false-color composite image that consists of smoke reflectance index, mid-infrared reflectance and water index was proposed to discriminate smoke from clouds. The smoke was shown in reddish color and the clouds were in pink-white. In order to create this false-color image, Smoke Aerosol Reflectance Index (SARI), Land Water Reflectance Index (LWRI) and Water Index (WI) were developed.

The method was applied to a MODIS scene of Russia and Mongolian forest fire in 2009. The false-color composite image showed smoke clearly in reddish color. Total smoke pixels were detected by their training dataset and the result was compared to those of multi-threshold method. Consequently, the proposed method detected smoke pixels more accurately than the multi-threshold method.

6. REFERENCES

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